

VectorCam Assembly Instructions

Document Details

Version 1.0

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1.0 Printed Components

Tools required: needle-nose pliers, utility blade.

1.1 Component Checklist

Locate all 3D-printed components required for the assembly of one device:

- 1 VectorCam Body
- 1 VectorCam Lid
- 1 Body Insert
- 2 LED Frame
- 2 Mosquito Slide
- 1 Eppendorf Slide

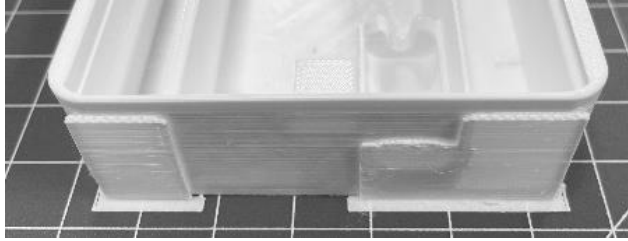


1.2 VectorCam Body

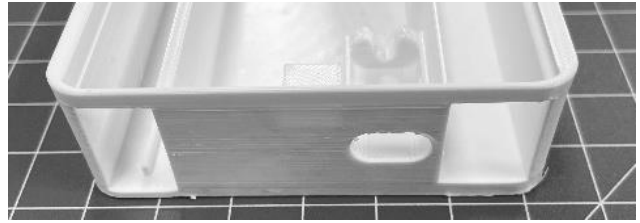
Remove any remaining support material and print overhangs in the following locations:

- bottom-side
- USB-C window
- accessory windows

BAD — support material remaining



GOOD — no supports, clean edges

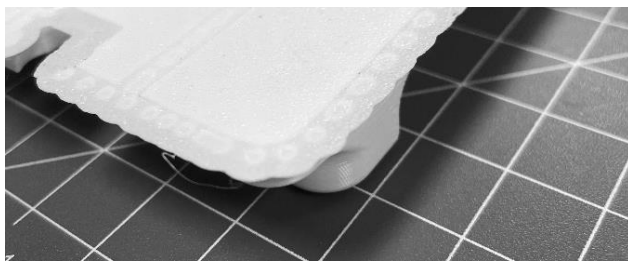


1.3 VectorCam Lid

Remove any remaining support material and print overhangs in the following locations:

- bottom-side
- side rails
- phone retainers

BAD — support material remaining



GOOD — no supports, clean edges

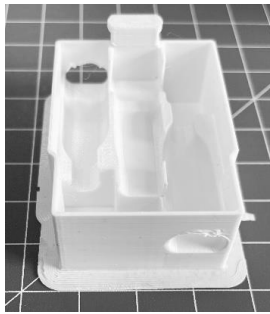


1.4 Body Insert

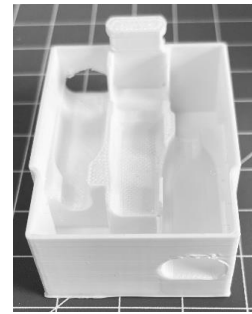
Remove any remaining support material and print overhangs in the following locations:

- USB-C windows
- base

BAD — support material remaining



GOOD — no supports, clean edges

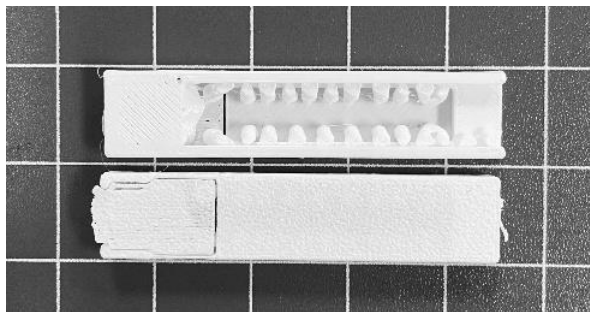


1.5 LED Frames

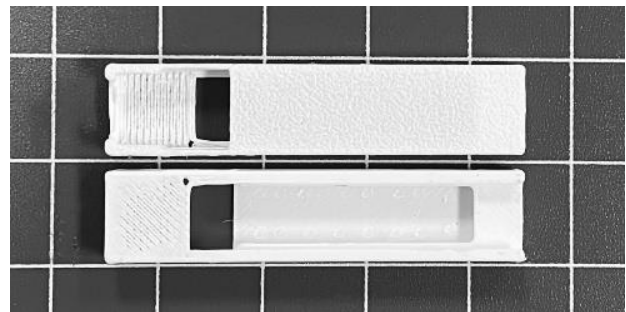
Remove any remaining support material and print overhangs in the following locations:

- bottom-side
- diffuser rails

BAD — support material remaining



GOOD — no supports, clean edges

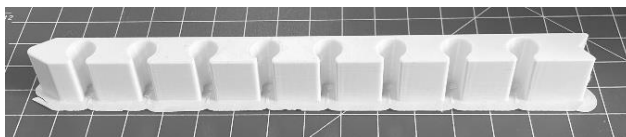


1.6 Mosquito Slides

Remove any remaining support material and print overhangs in the following locations:

- edges

BAD — support material remaining



GOOD — no supports, clean edges



1.7 Eppendorf Slide

Remove any remaining support material and print overhangs in the following locations:

- edges

BAD — support material remaining



GOOD — no supports, clean edges



2.0 Electronic and Optical Assembly

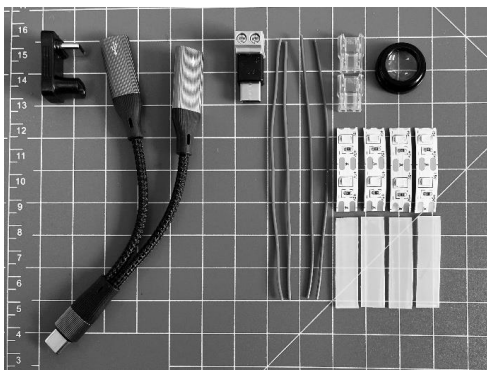
Tools required: needle-nose pliers, wire strippers, small Phillips-head screwdriver, USB-C power source.

2.1 Component Checklist

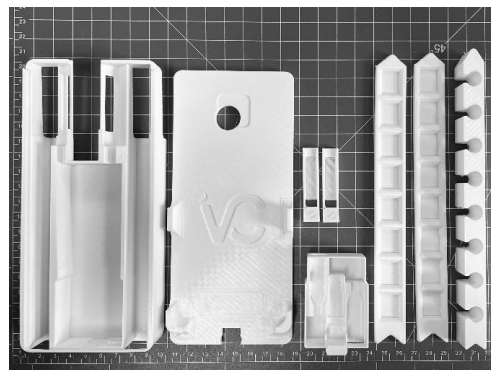
Locate all electronic, optical, and 3D-printed components required for the assembly of one device:

- 1 USB-C adapter (180°)
- 1 USB-C splitter
- 1 USB-C solderless plug (5V)
- 2 red insulated wires (100mm)
- 2 green insulated wires (100mm)
- 2 solderless LED adapters
- 4 LED strips (2 nodes/strip)
- 4 diffusers (9mm x 35mm)
- 15x lens
- 1 set of 3D-printed device components

Electronic and optical components



3D-printed device components



2.2 Power Connector Assembly

- Connect the female USB-C adapter to the male USB-C splitter, with the power input on the right and USB-C on the left.
- Insert the USB-C adapter and splitter assembly into the body insert, power input on the right and USB-C on the left.

- Insert the body insert into the VectorCam Body.

USB-C adapter connected to splitter

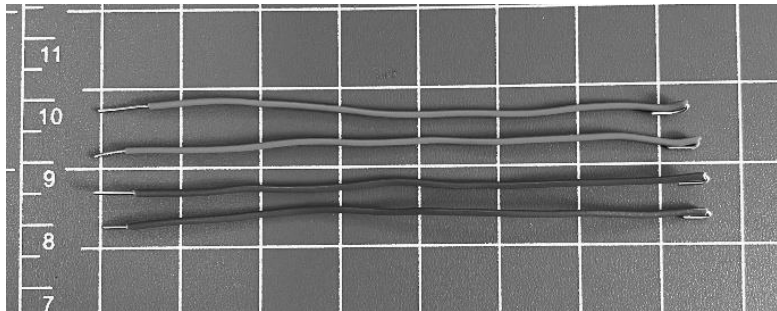


Adapter and splitter inserted into body insert



2.3 Wire Preparation

- Strip 5mm from both ends of each wire using wire strippers.
- Fold one end of each wire 180° using needle-nose pliers.

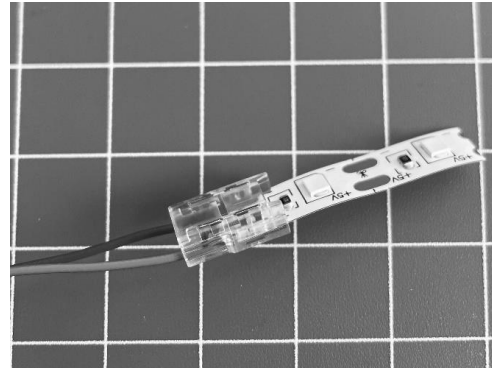
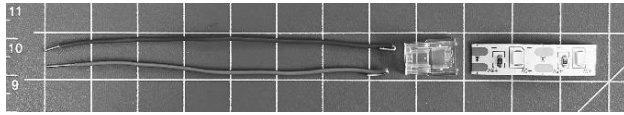


2.4 LED Strip Wiring

- Connect the wires to the lead end of the LED strip: red wire (+) and green wire (–) inside the LED adapter.
- Completely compress the LED adapter using needle-nose pliers.
- Repeat with the second LED strip, LED adapter, red wire, and green wire.

Wire connected at LED strip lead end

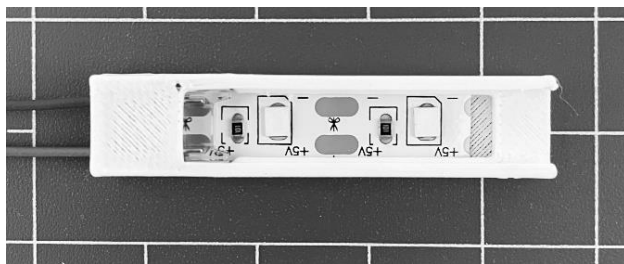
LED adapter compressed with pliers



2.5 LED Frame Assembly

- Insert both LED strips into the LED frames and pinch the LED connector flush.
- Slide the diffusers into the LED frames.
- Verify that the diffusers are flush with the LED frames and do not fall short or extend too far.
- Verify that the diffusers are secure by holding the frame and shaking the wires.

LED strip installed in frame (1 of 2)



LED strip installed in frame (2 of 2)

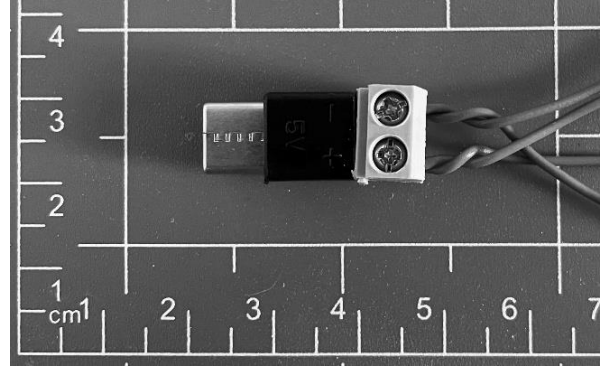
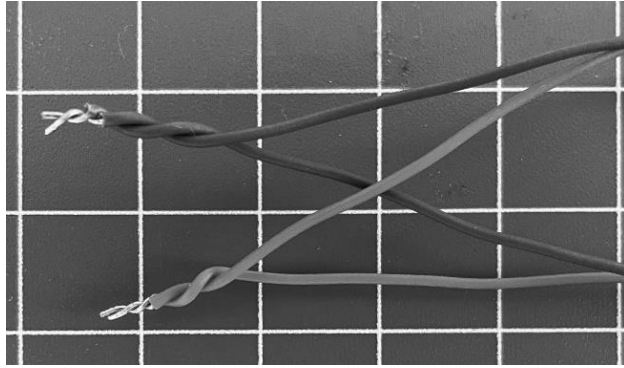


2.6 Solderless Plug Wiring

- Splice the red and green wires from each LED frame together using needle-nose pliers.
- Loosen both screws of the USB-C solderless plug.
- Fasten the spliced red (+) and green (–) wires in the USB-C solderless plug using the Phillips-head screwdriver.

Wires spliced together

Wires fastened in USB-C solderless plug

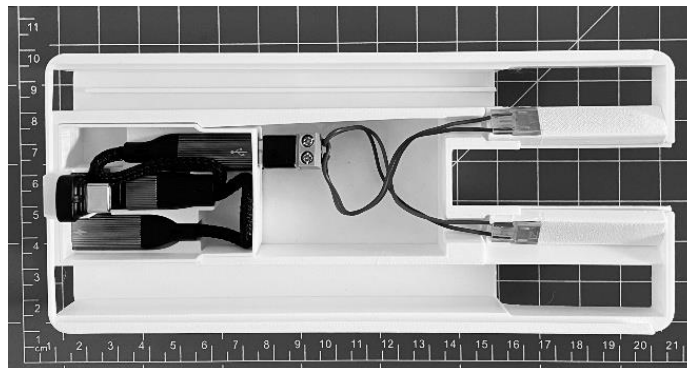


2.7 Wiring Verification

- Verify that the wires are securely attached to the LED strips by pulling on the connection.
- Verify that the wires are secure in the solderless plug by pulling on the connection.

2.8 Housing Assembly

- Insert the LED strips and frames into the VectorCam Body.
- Verify that the LED strips and frames are held in the VectorCam Body by tension.
- Connect the solderless plug to the body insert and USB-C assembly.



3.0 Final Assembly and Verification

3.1 Lens Installation

- Using a level surface, set the 15x lens into the VectorCam Lid.
- Verify that no visible gaps exist between the lid and the lens.



BAD — visible gap



GOOD — no gap



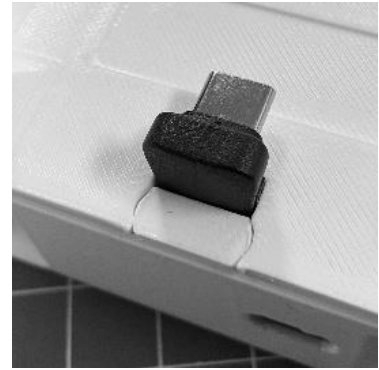
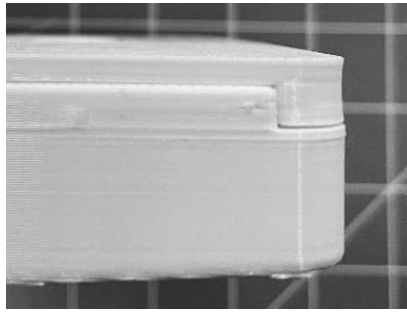
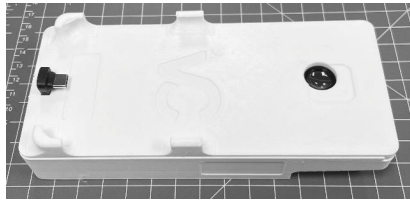
3.2 Lid and Body Assembly

- Slide the assembled VectorCam Lid into the assembled VectorCam Body.
- Verify that the interface between the lid and body is flush.
- Verify that the lid, USB-C adapter, and body are flush.

GOOD — flush overview

GOOD — flush side edge

GOOD — flush USB-C corner



3.3 Final Verification

- Verify that all 4 LED nodes turn on when the complete device is connected to a USB-C power source.

GOOD — LED check

